

Beautifframe

Beautiful Theorem-Like Environments for Typst

Version 0.4.5

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Contents

1	Introduction	4
1.1	Features	4
1.2	Quick Start	4
1.2.1	French Math Quick Start	4
2	Environments	5
2.1	Available Environments	5
2.2	Basic Usage	5
2.3	Numbering Control	6
2.3.1	Automatic Numbering (default)	6
2.3.2	No Numbering	6
2.3.3	Custom Numbering	6
2.3.4	Numbered Remarks	6
2.3.5	Section-Linked Numbering (LaTeX <code>\numberwithin</code> style)	6
2.4	References	7
2.4.1	Aliases	7
2.4.2	Counter Reset	7
3	Styles	8
3.1	Classic Style	8
3.2	Modern Style	8
3.3	Elegant Style	8
3.4	Colorful Style	9
3.5	Boxed Style	9
3.6	Minimal Style	10
3.7	Academic Style	10
3.8	BW Style	10
3.9	Cours Style	11
4	Variants	12
4.1	Assigning Variants	12
4.2	Setting All Variants at Once	12
4.3	Visual Gallery: All Variants $\times$ All Styles	13
4.3.1	Classic Style	14
4.3.2	Modern Style	15
4.3.3	Elegant Style	16
4.3.4	Colorful Style	17
4.3.5	Boxed Style	18
4.3.6	Minimal Style	19
4.3.7	Academic Style	20
4.3.8	BW Style	21
4.3.9	Cours Style	22
5	Proofs and QED Symbols	23
5.1	Basic Proof	23
5.2	QED Symbol Presets	23
5.3	Custom QED Symbol	23
6	Colors and Themes	24
6.1	Global Colors	24
6.2	Per-Environment Colors	24
6.3	Label Ink	24
6.4	Background Tints	25

6.5	Color Themes	26
6.6	Print-Friendly Modes	26
7	Language Presets	28
7.1	English	28
7.2	French	28
7.3	German	28
7.4	Spanish	28
7.5	Custom Labels	28
8	Layout Configuration	29
8.1	Classic Style Layout	29
8.2	Spacing	29
8.3	Typography	29
8.4	Boxed Style Options	29
9	Custom Environments	29
9.1	Parameters	30
9.2	Plural Forms	30
9.3	Resetting Custom Counters	31
10	French Math Preset	31
10.1	Preset Functions	31
10.2	French Environments	31
10.3	Challenge Callout (<code>defi</code>)	31
10.4	Formula Recap (<code>formule-end</code> / <code>formules-recap</code>)	32
10.5	Course Meta-Environments	32
10.6	Example	33
11	Instructor Mode	35
11.1	Worked Exercises	35
11.2	Instructor-Only Environments	35
12	QR Sidebar	36
12.1	Configuration	36
12.2	Usage	36
13	Student Fill Space	37
13.1	Parameters	37
13.2	Examples	37
14	Header Layout	38
14.1	Abbreviations	38
15	Trous (Fill-in Blanks)	39
15.1	Usage	39
15.2	Parameters	39
15.3	Configuration	39
15.4	The Two Builds Side by Side	40
15.5	Per-Style Rendering	40
16	API Reference	40
16.1	Environment Functions	40
16.2	Setup Function	41
16.3	Utility Functions	43

1 Introduction

Beautiframe is a Typst package for creating beautiful theorem-like environments (theorems, definitions, lemmas, proofs, etc.) with multiple visual styles.

1.1 Features

- **9 distinct styles:** classic, modern, elegant, colorful, boxed, minimal, academic, **bw**, **cours**
- **6 variants per style:** prominent, standard, subtle, accent, minimal, inline
- **Flexible mapping:** Assign any variant to any environment type
- **Independent counters:** Each environment type has its own counter
- **Customizable labels:** Change “Theorem” to “Théorème”, “Satz”, etc.
- **QED symbol presets:** □, ■, ■, CQFD,
- **Color themes:** Pre-built themes (ocean, forest, sunset, lavender)
- **Language presets:** French, German, Spanish
- **French Math Preset:** one-call setup for French secondary math courses
- **QR sidebar:** attach a QR code column to any environment
- **Environment references:** label theorem-like blocks and link back to their page
- **Student fill space:** blank, ruled lines, or dot grid appended inside any environment
- **Trous:** fill-in blanks that hold the content of the instructor build
- **Print-friendly modes:** color, grayscale, black & white

1.2 Quick Start

```
#import "@preview/beautiframe:0.4.5": *

#theorem(name: "Pythagorean"){
  In a right triangle:  $a^2 + b^2 = c^2$ 
}

#definition[
  A *limit* is the value that a function approaches.
]

#proof[
  The proof is left as an exercise.
]
```

1.2.1 French Math Quick Start

```
#import "@preview/beautiframe:0.4.5": *

#preset-french-math() // or #preset-french-math-bw()

#theoreme(name: "Pythagore")[Dans un triangle rectangle:  $a^2 + b^2 = c^2$ ]
#definitionfr[Une fonction continue préserve les limites.]
#pratique(space: "lines", space-height: 3cm)[Calculer la dérivée de  $f(x) = x^3$ .]
```

2 Environments

2.1 Available Environments

Beautiframed provides 8 environment types:

Environment	Default Variant	Counter	Usage
theorem	prominent	Optional	Main results
definition	standard	Optional	Foundational concepts
lemma	standard	Optional	Supporting results
proposition	standard	Optional	Secondary results
corollary	standard	Optional	Consequences
remark	subtle	Optional	Commentary
example	accent	Optional	Illustrations
proof	(special)	No	Demonstrations

2.2 Basic Usage

Each environment function accepts:

- **name:** Optional name (e.g., "Pythagorean" or [Pythagorean])
- **title:** Synonym for **name** — both are accepted, **name** takes priority
- **number:** auto (default), none, or custom value
- **label:** Typst label for `env-ref`, for example `<thm-pythagore>`
- **qr:** URL string to attach a QR code sidebar (requires `qr-renderer` configured)
- **space:** "empty", "lines", or "grid" — adds fill space for student work
- **space-height:** Height of the fill area (default 3cm)
- **body:** The content

```
#theorem(name: "Fermat's Last")[
  There are no positive integers  $a$ ,  $b$ ,  $c$  such that
   $a^n + b^n = c^n$  for  $n > 2$ .
]
```

```
// title: is accepted as a synonym for name:
#theorem(title: "Fermat's Last")[
  There are no positive integers  $a$ ,  $b$ ,  $c$  such that
   $a^n + b^n = c^n$  for  $n > 2$ .
]
```

Important: When the name contains math, use content syntax [...] instead of string "...", because strings do not process math markup:

```
// Wrong –  $f$  appears as literal text
#theorem(name: "Continuity of  $f$ ")[...]

// Correct
#theorem(name: [ $f$  is continuous]) [...]
```

Theorem 1 | There are no positive integers a , b , c such that $a^n + b^n = c^n$ for $n > 2$.
Fermat's Last

2.3 Numbering Control

2.3.1 Automatic Numbering (default)

```
#theorem[First theorem] // Theorem 1
#theorem[Second theorem] // Theorem 2
```

2.3.2 No Numbering

```
#theorem(number: none)[Unnumbered theorem]
#definition(number: none)[Unnumbered definition]
```

Theorem 1 | This theorem has no number.

Example

2.3.3 Custom Numbering

```
#theorem(number: "A")[Special theorem A]
#theorem(number: "★")[Starred theorem]
```

Theorem A | Special theorem A

2.3.4 Numbered Remarks

By default, remarks have no number. To number them:

```
#remark(number: auto)[This remark will be numbered.]
```

Remark 1 | This remark is numbered.

Remark | This remark is not numbered (default).

2.3.5 Section-Linked Numbering (LaTeX `\numberwithin` style)

Off by default. Two independent, opt-in settings control it:

- `link-to-section` — prefixes each environment number with the heading number. `false` (default) gives “Theorem 3”; `true` prefixes one heading level (“Theorem 2.3”); an integer N prefixes the first N heading levels (“Theorem 2.1.3” for `link-to-section: 2`).
- `counter-reset: "section"` — restarts every environment counter after each heading up to the `link-to-section` depth (level 1 when `link-to-section` is off). The default “manual” keeps counting across the whole document (reset by hand with `beautiframing-reset()`).

Combine both for the classic LaTeX behaviour:

```
#set heading(numbering: "1.1.")
#beautiframing-setup(link-to-section: true, counter-reset: "section")
```

= First section

```
#theorem[...] // Theorem 1.1
#theorem[...] // Theorem 1.2
```

= Second section

```
#theorem[...] // Theorem 2.1
```

```
#beautiframing-setup(link-to-section: 2)
```

== A subsection

```
#theorem[...] // Theorem 2.1.1
```

Both settings also apply to custom environments created with `new-env` (and hence to `formule`, `methode`, `pratique`, ...): with the settings above, `#formule[...]` renders as “Formule 1.1”. Refer-

ences made with `env-ref` / `env-refs` display the same section-linked number as the environment itself.

2.4 References

Add a Typst label to any environment, then reference it with `#env-ref(<label>)`. The reference text includes the environment label, number, and target page, and the whole text links to the labelled block.

```
#theorem(label: <thm-pythagore>, title: "Pythagore")[
  Dans un triangle rectangle:  $a^2 + b^2 = c^2$ .
]
```

```
Voir #env-ref(<thm-pythagore>).
// -> Théorème 1 (p. 3)
```

```
#remark(label: <rem-unites>)[Attention aux unités.]
Voir #env-ref(<rem-unites>).
// -> Remark (p. 3)
```

Use `page: false` to hide the page number:

```
#env-ref(<thm-pythagore>, page: false)
```

Use `page-style: "comma"` when the reference already sits inside parentheses:

```
(voir #env-ref(<thm-pythagore>, page-style: "comma"))
```

Use `env-refs` for several environments. Consecutive references with the same label are compacted:

```
#pratique(label: <prac-3>)[...]
#pratique(label: <prac-4>)[...]
#pratique(label: <prac-5>)[...]
#pratique(label: <prac-6>)[...]
```

```
Voir #env-refs(<prac-3>, <prac-4>, <prac-5>, <prac-6>, page: false).
// -> En pratique 3-6
```

```
Voir #env-refs(<prac-3>, <prac-4>, <prac-5>, <prac-6>, page-style: "comma").
// -> En pratique 3-6, pp. 4-5
```

```
#definition(label: <def-limite>)[...]
#proposition(label: <prop-limite>)[...]
```

```
Voir #env-refs(<def-limite>, <prop-limite>, page: false).
// -> Définition 1 et Proposition 2
```

2.4.1 Aliases

`envref` and `envrefs` are shorter spellings of `env-ref` and `env-refs`; they take the same arguments.

2.4.2 Counter Reset

Reset all counters manually:

```
#beautiframe-reset()
```

3 Styles

Beautiframe includes 9 visual styles. Each style provides 6 variants.

3.1 Classic Style

Traditional textbook layout with badge on the left and content with left border. Based on the exercise-bank pattern.

```
#beautiframe-setup(style: "classic")
#theorem(name: "Classic")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
#remark[This is the default style.]
```

Theorem 1 | In a right triangle: $a^2 + b^2 = c^2$
Classic

Definition 1 | A continuous function preserves limits.

Remark | This is the default style.

3.2 Modern Style

Contemporary design with thick accent bars and rule separators.

```
#beautiframe-setup(style: "modern")
#theorem(name: "Modern")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

Theorem 1 — *Modern*

In a right triangle: $a^2 + b^2 = c^2$

Definition 1

A continuous function preserves limits.

3.3 Elegant Style

Sophisticated styling with centered headers and decorative ornaments.

```
#beautiframe-setup(style: "elegant")
#theorem(name: "Elegant")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

◇ **THEOREM 1** ◇

(Elegant)

In a right triangle: $a^2 + b^2 = c^2$

• **DEFINITION 1** •

A continuous function preserves limits.

3.4 Colorful Style

Each environment type has a distinct color for visual differentiation.

```
#beautiframe-setup(style: "colorful")
#theorem[Red for theorems.]
#definition[Blue for definitions.]
#example[Green for examples.]
```

Theorem 1

Red for theorems.

Definition 1

Blue for definitions.

Example 1

Green for examples.

3.5 Boxed Style

Full framed boxes with optional backgrounds and header bars.

```
#beautiframe-setup(style: "boxed", border-radius: 3pt)
#theorem(name: "Boxed")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

Theorem 1 (*Boxed*)

In a right triangle: $a^2 + b^2 = c^2$

Definition 1

A continuous function preserves limits.

3.6 Minimal Style

Ultra-clean, print-friendly. Minimal ink usage.

```
#beautiframed-setup(style: "minimal")
#theorem(name: "Minimal") [In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition [A continuous function preserves limits.]
```

Theorem 1 (*Minimal*).

In a right triangle: $a^2 + b^2 = c^2$

Definition 1. A continuous function preserves limits.

3.7 Academic Style

Formal research paper style matching AMS/journal conventions.

```
#beautiframed-setup(style: "academic")
#theorem(name: "Academic") [In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition [A continuous function preserves limits.]
```

THEOREM 1 (*Academic*).

In a right triangle: $a^2 + b^2 = c^2$

Definition 1.

A continuous function preserves limits.

3.8 BW Style

Black-and-white two-column layout for printed French math courses (Gymnomath / cours Collège). A 3.35 cm right-aligned label column is separated from the content column by a horizontal rule (standard variant) or a box border (boxed/prominent variants).

Variants: **standard** (side-block with rule), **boxed** (light rect), **prominent** (thicker rect, for théorèmes), **accent** (env-color stroke and label), **minimal** (inline label), **inline**, **proof** (with QED).

```
#preset-french-math-bw()
#theoreme(name: "Pythagore") [Dans un triangle rectangle:  $a^2 + b^2 = c^2$ ]
#definitionfr [Une fonction continue préserve les limites.]
#remarque [La réciproque est généralement fausse.]
#preuve [Par application directe de la définition.]
```

Théorème 1
Pythagore

Dans un triangle rectangle: $a^2 + b^2 = c^2$

Définition 1 Une fonction continue préserve les limites.

Remarque La réciproque est généralement fausse.

Preuve. Par application directe de la définition. □

3.9 Cours Style

French course style with a 2 cm label column that overhangs 1 cm into the left margin, and content framed with a thin left border. Optimised for A4 course sheets with 2.2 cm margins. Blue accent color by default.

Variants: **standard** (accent-colored border and label), **accent** (per-env color), **subtle** (muted), **minimal** (inline, for remarques), **inline**, **proof** (with QED).

```
#preset-french-math()
#theoreme(name: "Valeurs intermédiaires")[
  Si  $f$  est continue sur  $[a ; b]$ , pour tout  $y$  entre  $f(a)$  et  $f(b)$  il existe
   $c$  tel que  $f(c) = y$ .
]
#definitionfr[Une suite converge si elle admet une limite finie.]
#remarque[La réciproque est fausse en général.]
```

Théorème 1 | Si f est continue sur $[a; b]$, pour tout y entre $f(a)$ et $f(b)$ il existe c tel que $f(c) = y$.
Valeurs intermédiaires

Définition 1 | Une suite converge si elle admet une limite finie.

Remarque La réciproque est fausse en général.

4 Variants

Each style provides variants that can be assigned to any environment type. The 6 core variants are available in all styles:

Variant	Description	Default for
prominent	Strongest visual emphasis, thick borders, bold colors	theorem
standard	Normal, balanced styling	definition, lemma, proposition, corollary
subtle	Lighter, less prominent, muted colors	remark
accent	Uses environment-specific color	example
minimal	Very light styling, minimal visual elements	—
inline	No structural elements, flows with text	—

Boxed style has 3 additional variants:

Variant	Description
titled	Label breaks the top border line (beautitled-style)
centered	Label centered at top, breaking the border
corner	L border (top + left sides)
corner2	Inverted L border (left + bottom sides)

4.1 Assigning Variants

Map any variant to any environment type:

```
#beautifframe-setup(
  theorem-variant: "prominent", // Theorems get maximum emphasis
  lemma-variant: "subtle",      // Lemmas are de-emphasized
  remark-variant: "inline",     // Remarks flow inline
  example-variant: "accent",    // Examples use their color
)
```

4.2 Setting All Variants at Once

The `default-variant` parameter sets **all** 7 environment variants in a single call. Individual overrides are applied after, so they win:

```
// Make every environment use the boxed variant
#beautifframe-setup(default-variant: "boxed")

// All boxed, except theorems which get prominent
#beautifframe-setup(default-variant: "boxed", theorem-variant: "prominent")
```

This is useful when switching styles (e.g., `bw` → `cours`) and wanting a uniform appearance, or when preparing a worksheet where all environments should have the same fill-space variant.

All minimal via default-variant:

Theorem 1 Content.

Sample

Definition 1 Content.

Remark Content.

All standard, theorem overridden to prominent:

Theorem 1 | Content.

Sample

Definition 1 | Content.

Remark | Content.

4.3 Visual Gallery: All Variants × All Styles

4.3.1 Classic Style

Prominent:

Theorem 1 | Sample text.
Name

Subtle:

Theorem 3 | Sample text.
Name

Minimal:

Theorem 5 Sample text.
Name

Standard:

Theorem 2 | Sample text.
Name

Accent:

Theorem 4 | Sample text.
Name

Inline:

Theorem 6 (*Name*). Sample text.

4.3.2 Modern Style

Prominent:

Theorem 1 — *Name*

Sample text.

Subtle:

Theorem 3 — *Name*

Sample text.

Minimal:

Theorem 5 — *Name*

Sample text.

Standard:

Theorem 2 — *Name*

Sample text.

Accent:

Theorem 4 — *Name*

Sample text.

Inline:

Theorem 6 — *Name* — Sample text.

4.3.3 Elegant Style

Prominent:

◇

THEOREM 1

◇

(Name)

Sample text.

Standard:

•

THEOREM 2

•

(Name)

Sample text.

Subtle:

Theorem 3

(Name).

Sample text.

Accent:

★

THEOREM 4

★

(Name)

Sample text.

Minimal:

THEOREM 5

(Name)

Sample text.

Inline:

THEOREM 6

(Name).

Sample text.

4.3.4 Colorful Style

Prominent:

Theorem 1 (*Name*)

Sample text.

Subtle:

Theorem 3 (*Name*)

Sample text.

Minimal:

Theorem 5 (*Name*)

Sample text.

Standard:

Theorem 2 (*Name*)

Sample text.

Accent:

Theorem 4 (*Name*)

Sample text.

Inline:

Theorem 6 (*Name*). Sample text.

4.3.5 Boxed Style

Prominent:

Theorem 1 (*Name*)

Sample text.

Standard:

Theorem 2 (*Name*)

Sample text.

Subtle:

Theorem 3 (*Name*)

Sample text.

Accent:

Theorem 4 (*Name*)

Sample text.

Minimal:

Theorem 5 (*Name*)

Sample text.

Inline:

[Theorem 6: *Name*] Sample text.

Titled:

Theorem 7 (*Name*)

Sample text.

Centered:

Theorem 8 (*Name*)

Sample text.

Corner:

Theorem 9 (*Name*)

Sample text.

Corner2:

Theorem 10 (*Name*)

Sample text.

4.3.6 Minimal Style

Prominent:

Theorem 1 (*Name*).

Sample text.

Subtle:

Theorem 3 (*Name*). Sample text.

Minimal:

Theorem 5. Sample text.

Standard:

Theorem 2 (*Name*). Sample text.

Accent:

THEOREM 4 (*Name*). Sample text.

Inline:

(**Theorem 6:** *Name*) Sample text.

4.3.7 Academic Style

Prominent:

THEOREM 1 (*Name*).

Sample text.

Subtle:

Theorem 3 (*Name*). Sample text.

Minimal:

Theorem 5. Sample text.

Standard:

Theorem 2 (*Name*).

Sample text.

Accent:

THEOREM 4 (*Name*).

Sample text.

Inline:

Theorem 6 (*Name*): Sample text.

4.3.8 BW Style

Prominent:

Théorème 1

Name

Sample text.

Standard:

Théorème 2

Name

Sample text.

Boxed:

Théorème 3

Name

Sample text.

Accent:

Théorème 4

Name

Sample text.

Minimal:

Théorème 5

(Name)

Sample text.

Inline:

Théorème 6

(Name).

Sample text.

4.3.9 Cours Style

Standard:

Théorème 1 | Sample text.
Name

Subtle:

Théorème 3 | Sample text.
Name

Inline:

Théorème 5 (*Name*). Sample text.

Accent:

Théorème 2 | Sample text.
Name

Minimal:

Théorème 4 Sample text.
Name

5 Proofs and QED Symbols

5.1 Basic Proof

```
#proof[
```

```
    By direct calculation, we have  $2^2 = 4$ .
```

```
]
```

Preuve. By direct calculation, we have $2^2 = 4$.

□

5.2 QED Symbol Presets

Beautiframe provides several QED symbol presets:

```
#qed-square()      // □ (default)
#qed-filled()      // ■
#qed-tombstone()   // ■
#qed-cqfd()        // CQFD (French)
#qed-slashes()     // //
#qed-text()        // Q.E.D.
#qed-none()        // (no symbol)
```

Default (□):

Preuve. Hollow square.

□

Filled (■):

Preuve. Filled square.

■

Tombstone (■):

Preuve. Tombstone symbol.

■

CQFD:

Preuve. French style.

CQFD

Slashes:

Q.E.D.:

Preuve. Double slash.

//

Preuve. Latin abbreviation.

Q.E.D.

5.3 Custom QED Symbol

```
#beautiframe-setup(qed-symbol: text(fill: green, sym.checkmark))
```

Preuve. Custom green checkmark.

✓

6 Colors and Themes

6.1 Global Colors

```
#beautiframing-setup(
  primary-color: rgb("#1a5276"),
  secondary-color: rgb("#7f8c8d"),
  accent-color: rgb("#2980b9"),
)
```

6.2 Per-Environment Colors

```
#beautiframing-setup(
  theorem-color: rgb("#c0392b"), // Red
  definition-color: rgb("#2980b9"), // Blue
  lemma-color: rgb("#8e44ad"), // Purple
  example-color: rgb("#27ae60"), // Green
  remark-color: rgb("#7f8c8d"), // Gray
)
```

By default only the accent variants paint with these colors: every other variant uses the single `accent-color`, so a document keeps one uniform hue. Set `env-colors: true` to make **all** variants of **all** styles follow the per-environment colors, including the boxed-only variants (`titled`, `centered`, `corner`, `corner2`):

```
#beautiframing-setup(env-colors: true)
```

Default (`env-colors: false`) — one accent hue for every environment:

Théorème

Every frame uses `accent-color`.

Exemple

Including this one.

With `env-colors: true` — each environment paints with its own colour:

Théorème

Red, from `theorem-color`.

Exemple

Green, from `example-color`.

6.3 Label Ink

`label-color` decides what the header text is painted with:


```
#beautiframe-setup(label-color: auto)      // each style's own choice (default)
#beautiframe-setup(label-color: "base")    // follow the environment colour
#beautiframe-setup(label-color: red)       // one explicit colour everywhere
```

"base" combined with `env-colors: true` gives a header written in the same colour as its frame, which pairs well with `header-layout: "prefix"`:

```
#beautiframe-setup(
  env-colors: true, label-color: "base",
  header-layout: "prefix", label-abbrev: true,
)
// → Thm 1: Théorème de Pythagore (title and frame both in the theorem colour)
```

Thm 1: Pythagore

In a right triangle: $a^2 + b^2 = c^2$.

Ex 1: A first check

$$3^2 + 4^2 = 5^2.$$

6.4 Background Tints

Filled boxes tint their colour towards white. A **fixed** lightening makes light hues vanish and dark hues shout: a yellow lightened by 92% is barely visible while a green still reads strongly. The default `background-tint: auto` therefore lightens each colour by however much it takes to reach `background-lightness`, so every tint reads with the same strength.

```
#beautiframe-setup(background-tint: auto)      // perceptual (default)
#beautiframe-setup(background-lightness: 0.93) // target lightness, 0..1
#beautiframe-setup(background-tint: 92%)       // fixed lightening for all hues
```

Lower `background-lightness` for stronger tints, raise it for paler ones.

Fixed lightening (`background-tint: 92%`) — the yellow all but disappears:

Théorème

Navy tint.

Remarque

Yellow tint.

Exemple

Green tint.

Perceptual (`background-tint: auto`, the default) — every tint reads alike:

Théorème

Navy tint.

Remarque

Yellow tint.

Exemple

Green tint.

6.5 Color Themes

Pre-built color schemes:

Ocean Theme:

Théorème

Blue tones throughout.

Forest Theme:

Théorème

Green tones throughout.

Sunset Theme:

Théorème

Warm red and orange tones.

Lavender Theme:

Théorème

Purple tones throughout.

6.6 Print-Friendly Modes

For B&W printing:

```
#beautiframe-setup(color-mode: "color") // Full color (default)
#beautiframe-setup(color-mode: "grayscale") // Grayscale
#beautiframe-setup(color-mode: "bw") // Pure black and white
```

Color mode:

Théorème
Full color styling.

Grayscale mode:

Théorème
Grayscale for B&W printers.

B&W mode:

Théorème
Pure black and white.

7 Language Presets

7.1 English

`preset-english()` restores the built-in English labels, plurals and abbreviations. It is the way back after `preset-french()`, `preset-german()`, `preset-spanish()` or `preset-french-math()`.

`#preset-english()`

7.2 French

`#preset-french()`

Théorème 1 | Dans un triangle rectangle: $a^2 + b^2 = c^2$
Pythagore

Définition 1 | Une fonction continue préserve les limites.

Preuve. Immédiat. □

7.3 German

`#preset-german()`

Satz 1 | In einem rechtwinkligen Dreieck: $a^2 + b^2 = c^2$
Pythagoras

Definition 1 | Eine stetige Funktion erhält Grenzwerte.

Beweis. Offensichtlich. □

7.4 Spanish

`#preset-spanish()`

Teorema 1 | En un triángulo rectángulo: $a^2 + b^2 = c^2$
Pitágoras

Definición 1 | Una función continua preserva límites.

Demostración. Inmediato. □

7.5 Custom Labels

```
#beautiframe-setup(
  theorem-label: "Théorème",
  definition-label: "Définition",
  proof-label: "Démonstration",
)
```

8 Layout Configuration

8.1 Classic Style Layout

The classic style uses a grid layout with configurable dimensions:

```
#beautiframing-setup(
  line-position: 2cm,    // Distance from left to vertical line
  label-extra: 1cm,     // Extension into left margin
  border-width: 1.5pt,  // Line thickness
)
```

8.2 Spacing

```
#beautiframing-setup(
  theorem-above: 1.2em, // Space before theorems
  theorem-below: 1em,   // Space after theorems
  header-gap: 0.4em,    // Gap between label and body
)
```

8.3 Typography

Control label appearance:

```
#beautiframing-setup(
  label-weight: "bold", // "bold", "regular", "semibold", etc.
  label-size: 11pt,     // Size of "Theorem", "Definition", etc.
  name-style: "italic", // Style for theorem names: "italic" or "normal"
)
```

Bold label (default):

Theorem 1 (*Name*)

Sample text.

Regular label:

Theorem 2 (*Name*)

Sample text.

8.4 Boxed Style Options

```
#beautiframing-setup(
  inset: (x: 1em, y: 0.8em), // Padding inside boxes
  border-radius: 4pt,         // Rounded corners
  border-width: 1.5pt,        // Border thickness
)
```

9 Custom Environments

Create your own environment types with independent counters using `new-env`:

```
// Create custom environments
#let conjecture = new-env("Conjecture", base: "theorem")
#let propriete = new-env("Propriété", base: "definition", numbered: false)
#let formule = new-env("Formule", base: "lemma", color: green)
```

```
#let axiom = new-env("Axiom", base: "theorem", numbered: true)

// Use them like built-in environments
#conjecture[Every even number greater than 2 is the sum of two primes.]
#conjecture(name: "Goldbach")[Famous unsolved problem.]
#propriete[A property without number.]
#formule[The quadratic formula:  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ ]
```

Conjecture 1

Every even number greater than 2 is the sum of two primes.

Conjecture 2 (*Goldbach*)

Famous unsolved problem in number theory.

Axiom 1

Two points determine a unique line.

9.1 Parameters

```
#let my-env = new-env(
  "Label",           // Display label (required)
  base: "theorem",   // Inherit style from: theorem, definition, lemma, etc.
  numbered: true,     // Auto-number by default
  color: none,        // Optional custom color
)
```

9.2 Plural Forms

All environments support a `plural` parameter. Default plurals are provided for English and all language presets:

```
#theorem(plural: true)[Multiple theorems grouped together.]
#definition(plural: true)[Several related definitions.]
```

Theorem 1

A single theorem.

Theorems 2

Multiple theorems can be grouped in one box.

Language presets automatically set the correct plural forms:

```
#preset-english() // Sets: Theorems, Definitions, Lemmas, etc.
#preset-french()   // Sets: Théorèmes, Définitions, Lemmes, etc.
#preset-german()   // Sets: Sätze, Definitionen, Lemmata, etc.
#preset-spanish()  // Sets: Teoremas, Definiciones, Lemas, etc.
```

Custom environments also support plurals:

```
#let conjecture = new-env("Conjecture", plural: "Conjectures", base: "theorem")
#conjecture(plural: true)[Two famous conjectures.]
```

9.3 Resetting Custom Counters

```
#reset-env("Conjecture") // Reset specific custom environment
#beautiframe-reset()      // Reset all built-in counters
```

10 French Math Preset

The French math preset configures beautiframe in a single call for French secondary and post-secondary math courses.

10.1 Preset Functions

```
// Color version – cours style, blue accent, bold labels, QED square
#preset-french-math()
```

```
// Black-and-white version – bw style, 8.4pt labels, luma palette
#preset-french-math-bw()
```

```
// Reset all counters including custom French environments
#beautiframe-reset-french-math()
```

10.2 French Environments

All French environments accept `name:` / `title:` (alias), `qr:`, `space:`, `space-height:`.

Function	Label	Base	Numbered
theoreme	Théorème	theorem	Yes
definitionfr	Définition	definition	Yes
propositionfr	Proposition	proposition	Yes
exemplefr / exemple	Exemple	example	Yes
remarque	Remarque	remark	No
corollaire	Corollaire	corollary	Yes
preuve	Preuve	proof	No
pratique	En pratique	example	Yes
propriete	Propriété	corollary	No
formule	Formule	lemma	Yes
formules(...)	Formules	lemma	Yes
methode	Méthode	proposition	Yes
regles	Règle	proposition	No
guided-example	Exemple guidé	example	Yes
notation(...)	Notation	remark	No
discussion(...)	Discussion	remark	No
objectifs / objectif	Objectifs d'apprentissage	lemma	No
concepts	Concepts clés	lemma	No
glossaire	Glossaire	lemma	No
defi / défi	 Défi	remark	No

10.3 Challenge Callout (defi)

`#defi` renders a compact, unnumbered challenge callout using the remark style. The icon and title are optional.

Parameter	Type	Default	Description
<code>title</code>	content or none	Optional title appended after “Défi :”	
<code>icon</code>	content or none	[🎯]	Decorative icon; set <code>none</code> to hide
<code>label</code>	label or none	<code>none</code>	Cross-reference label
<code>qr</code>	string or none	<code>none</code>	QR sidebar URL
<code>space</code>	string or none	<code>none</code>	Fill space type
<code>space-height</code>	length	3cm	Height of fill area
<code>body</code>	content	—	Challenge content

```
#defi[Résoudre  $x^2 - 5x + 6 = 0$ .]
#defi(title: [Y arrivez-vous ?])[Montrer que  $\sqrt{2}$  est irrationnel.]
#defi(icon: [🎯])[Justifier chaque étape du raisonnement.]
#defi(icon: none)[Sans icône.]
```

🎯 Défi Résoudre $x^2 - 5x + 6 = 0$.

🎯 Défi : Y arrivez-vous ? Montrer que $\sqrt{2}$ est irrationnel.

10.4 Formula Recap (formule-end / formules-recap)

Collect formulas throughout a chapter and print them all at once in a recap box at the end. Use `#formule-end` to register a formula (nothing is displayed at the call site), then `#formules-recap` to render the collected list.

```
// Register formulas anywhere in the chapter (nothing rendered in place)
#formule-end([Discriminant],  $\Delta = b^2 - 4ac$ )
#formule-end([Racines],  $x = \frac{-b \pm \sqrt{\Delta}}{2a}$ )

// Render the recap (clears the list by default)
#formules-recap()

// Custom title
#formules-recap(title: [Formules du chapitre])

// Keep the list for a second recap (clear: false)
#formules-recap(clear: false)

#recap-formules() // alias of formules-recap
```

Parameter	Type	Default	Description
<code>label</code>	content	—	Formula name displayed in bold
<code>formula</code>	content	—	The formula content

Parameter (formules-recap)	Type	Default	Description
<code>title</code>	content	[Formules à retenir]	Box header
<code>clear</code>	bool	true	Clear the collection after printing

Formules à retenir

Discriminant :

$$\Delta = b^2 - 4ac$$

Racines :

$$x_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a}$$

10.5 Course Meta-Environments

Three unnumbered environments intended for course structure:

- `#objectifs` / `#objectif` — learning objectives at the start of a chapter
- `#concepts` — key concepts summary
- `#glossaire` — glossary or vocabulary list


```
#objectifs[
  - Savoir résoudre une équation du second degré
  - Connaître la formule du discriminant
]

#concepts[Équation, discriminant, racine, trinôme.]

#glossaire[*Racine* : valeur de  $x$  pour laquelle  $f(x) = 0$ .]
```

Objectifs d'apprentissage | • Savoir résoudre une équation du second degré.
• Connaître et appliquer la formule du discriminant.

Concepts clés | Équation, discriminant, racine, trinôme du second degré.

10.6 Example

```
#import "@preview/beautiframe:0.4.5": *
#preset-french-math()

#theoreme(name: "Pythagore")[
  Dans tout triangle rectangle d'hypoténuse  $c$ :  $a^2 + b^2 = c^2$ .
]

#definitionfr[Une suite converge si elle admet une limite finie.]

#propriete[Toute suite monotone et bornée converge.]

#formule[
   $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ 
]

#pratique(space: "lines", space-height: 3cm)[
  Résoudre  $2x^2 - 5x + 3 = 0$ .
]

#preuve[Par calcul direct avec le discriminant.]
```

Théorème 1 | Dans tout triangle rectangle d'hypoténuse c : $a^2 + b^2 = c^2$.
Pythagore

Définition 1 | Une suite converge si elle admet une limite finie.

Propriété | Toute suite monotone et bornée converge.

Formule 1 |
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Notation On note f' la dérivée de f , et $f^{(n)}$ la dérivée n -ième.

Discussion La condition $a > 0$ est nécessaire mais pas suffisante.

En pratique 1 | Résoudre $2x^2 - 5x + 3 = 0$.

Preuve. Par calcul direct avec le discriminant.

□

11 Instructor Mode

A single `instructor-mode` switch turns one source file into two documents: the student handout (default) and the instructor version with corrections.

11.1 Worked Exercises

`worked-exercise` renders an exercise statement; its `correction:` block is shown only when `instructor-mode: true`:

```
#worked-exercise(
  title: "Dérivée d'un produit",
  correction: [On applique  $(u \cdot v)' = u'v + u \cdot v'$  ...],
)[
  Dériver  $f(x) = x^2 \sin(x)$ .
]
```

```
// In the instructor build:
#beautiframe-setup(instructor-mode: true)
```

Options:

- `correction-title`: overrides the heading of the correction block (default: the `correction-label` config, “Correction”).
- `correction-renderer`: config — a `(title, body) => content` function replacing the default framed block.
- `discussion` accepts the same `correction:` / `correction-title`: options.

11.2 Instructor-Only Environments

Every environment (built-ins and `new-env` customs) accepts `instructor: true` to hide the **entire block** from the student version:

```
#remark(instructor: true)[
  Insister sur le cas  $\Delta = 0$  — erreur fréquente au test.
]
```

12 QR Sidebar

Any environment can display a rendered QR code (or any content) in a right sidebar column.

12.1 Configuration

```
// Configure once in your preamble
#beautiframing-setup(
  qr-renderer: url => image.decode(
    tiaoma.qrcode(url, options: (border: 0)),
    format: "svg", width: 1.85cm
  ),
  qr-width: 1.85cm,
)
```

The `qr-renderer` key accepts a function `url => content`. When set to `none` (default), no sidebar is shown even if `qr:` is passed.

12.2 Usage

```
#theorem(qr: "https://example.com/proof")[
  In a right triangle:  $a^2 + b^2 = c^2$ .
]

#definitionfr(qr: "https://wiki.example.com/continuity")[
  Une fonction continue préserve les limites.
]
```

The QR sidebar works with all environments including custom ones created with `new-env`.

13 Student Fill Space

Append a blank fill area inside any environment for students to write their answers.

13.1 Parameters

- **space:** — "empty" (blank block), "lines" (8 mm ruled lines), "grid" (5 mm dot grid)
- **space-height:** — height of the fill area (default: 3cm)

13.2 Examples

```
// Blank area
```

```
#pratique(space: "empty", space-height: 2cm)[Define continuity.]
```

```
// Ruled lines
```

```
#pratique(space: "lines", space-height: 4cm)[Solve  $2x - 5 = 7$ .]
```

```
// Dot grid
```

```
#exemple(space: "grid", space-height: 5cm)[Sketch  $f(x) = x^2$ .]
```

Ruled lines (space: "lines", height 3 cm):

En pratique 1 Calculer la dérivée de $f(x) = 3x^2 - 2x + 1$.

Dot grid (space: "grid", height 3.5 cm):

Exemple 1 Représenter $f(x) = -x^2 + 4$ sur $[-3; 3]$.

14 Header Layout

By default the environment label leads and the title follows in parentheses: **Remark 2 (What is analysis?)**. When the title is the interesting half, swap the emphasis with `header-layout`.

Value	Renders
"label-first" (default)	Remark 2 (What is analysis?)
"title-first"	What is analysis? (Remark 2)
"prefix"	Rem 2: What is analysis?

```
#beautiframe-setup(
  header-layout: "title-first",    // or "prefix"
  label-abbrev: true,              // Remark → Rem, Theorem → Thm, ...
  prefix-separator: ":",          // used by "prefix"
)
```

The swap only applies to environments that **have** a title: `#definition[...]` without a name keeps rendering as **Definition 1**, since there is nothing to promote. Plural forms and custom `new-env` labels are never abbreviated.

Every style follows automatically, because the redistribution happens before the style is called: styles keep rendering their prominent half and their secondary half, and only the content of those halves changes. In `"prefix"` the demoted label is set at regular weight so the title carries the emphasis.

header-layout: "label-first" (default):

Remark The study of limits, in one word.
What is analysis?

header-layout: "title-first":

What is analysis? **The study of limits, in one word.**
Remark

header-layout: "prefix" with label-abbrev: true:

Rem: **What is analysis?** The study of limits, in one word.

A nameless environment is untouched by the layout:

Remark There is no title here to promote, so the label stays put.

14.1 Abbreviations

```
#beautiframe-setup(
  theorem-abbrev: "Thm", definition-abbrev: "Def", lemma-abbrev: "Lem",
  proposition-abbrev: "Prop", corollary-abbrev: "Cor",
  remark-abbrev: "Rem", example-abbrev: "Ex", proof-abbrev: "Pf",
)
```

`preset-french()` sets the French forms (Déf, Thm, Rem, Pr, ...) and a narrow non-breaking space before the colon of the `"prefix"` layout.

15 Trous (Fill-in Blanks)

A `#trou` is fill space that **carries content**. It prints as reserved, correctly sized blank space in the student build, and as the content itself in the instructor build. One source file therefore produces the sheet the class writes on and the sheet the teacher reads from.

The reserved height is **not** the height of the typeset content: a student's hand needs roughly twice the room that typeset text occupies, so the measured height is multiplied by `trou-scale` (default 2.0). With `fill: "lines"` the result then snaps up to a whole number of `trou-line-gap` rules, so three lines of typeset content reserve four ruled lines. An explicit `height` overrides all of this and is used as given.

Where `space:` appends anonymous blank space to an environment, a `trou` holds what the class is meant to produce there: the example, the counterexample, the answer, the sketch.

15.1 Usage

```
#trou[La suite $1/n$ tend vers $0$ sans jamais l'atteindre.]
```

```
#trou(hint: [contre-exemple])[La fonction de Dirichlet.]
```

```
#trou(fill: "lines", height: 4cm)[Esquisse du graphe.]
```

Une fonction `#trou-inline[continue]` sur `[$[a; b]$]` atteint ses bornes.

The instructor build is one switch away:

```
#beautiframe-setup(instructor-mode: true)
```

15.2 Parameters

Parameter	Default	Meaning
height	auto	auto measures the hidden content and scales it; or give a fixed length, used as is
scale	auto	Handwriting factor on the measured height (falls back to <code>trou-scale</code>)
fill	auto	"empty", "lines" or "grid" (falls back to <code>trou-fill</code>)
frame	auto	Frame the reserved area (falls back to <code>trou-frame</code>)
hint	none	Short cue printed in the student build
min-height	auto	Floor for the reserved height (falls back to <code>trou-min-height</code>)
padding	auto	Breathing room added to the measured height

`#trou-inline(width: auto, body)` blanks a single word or expression inside a printed sentence, sized to the hidden content unless `width` is given.

15.3 Configuration

```
#beautiframe-setup(
  trou-fill: "empty",           // "empty" | "lines" | "grid"
  trou-scale: 2.0,              // handwriting factor on the measured height
  trou-line-gap: 8mm,           // ruling pitch, and the quantum "lines" snaps to
  trou-frame: true,             // frame the reserved space
  trou-color: luma(65%),         // ink of rules and frames (student build)
  trou-padding: 0.6em,          // added to the measured height
  trou-min-height: 1cm,         // never reserve less than this
  trou-max-height: none,        // cap the reserved height
```

```

    trou-hint-size: 8pt,           // size of the hint text
    trou-mark-instructor: true,   // flag filled content in the instructor build
    trou-mark-color: none,        // none = accent-color
)

```

15.4 The Two Builds Side by Side

Student build (`instructor-mode: false`):

un contre-exemple

Une fonction _____ sur $[a; b]$ y atteint ses bornes.

Instructor build (`instructor-mode: true`) — same source, content printed:

un contre-exemple

La fonction de Dirichlet.

On pose $u_n = \frac{1}{n}$. La suite décroît vers 0 sans jamais l'atteindre.

Une fonction continue sur $[a; b]$ y atteint ses bornes.

A trou also works inside an environment body, where it drops the label-column layout because that column is already taken:

Example 1

Suite convergente

Donner une suite qui converge vers 0 sans jamais l'atteindre.

15.5 Per-Style Rendering

Every style renders trous in its own visual language: `bw` and `cours` and `classic` put the hint in their label column and the reserved space in the content column, `boxed` breaks the hint through the top border, `modern` and `colorful` keep their left bar, `elegant` sets the space between two rules, `minimal` closes it with a single baseline rule, and `academic` uses a thin frame. A trou placed **inside** an environment body automatically drops the label-column layout, since that column is already occupied.

16 API Reference

16.1 Environment Functions

```

// Built-in environments (all accept name:/title:, number:, qr:, space:, space-
height:)
#theorem(name: none, title: none, number: auto, qr: none, space: none, space-height:
3cm)[body]

```



```

#definition(name: none, title: none, number: auto, qr: none, space: none, space-
height: 3cm)[body]
#lemma(name: none, title: none, number: auto, qr: none, space: none, space-height:
3cm)[body]
#proposition(name: none, title: none, number: auto, qr: none, space: none, space-
height: 3cm)[body]
#corollary(name: none, title: none, number: auto, qr: none, space: none, space-
height: 3cm)[body]
#remark(name: none, title: none, number: none, qr: none, space: none, space-height:
3cm)[body]
#example(name: none, title: none, number: auto, qr: none, space: none, space-height:
3cm)[body]
#proof[body]

// French aliases
#theoreme      = theorem
#definitionfr  = definition
#propositionfr = proposition
#exemplefr     = exemple = example
#remarque      = remark
#corollaire    = corollary
#preuve(body)  = proof

// French custom environments
#pratique(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#propriete(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#formule(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#formules(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#methode(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#notation(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#discussion(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]

// Trous – reserved space in the student build, content in the instructor build
#trou(height: auto, scale: auto, fill: auto, frame: auto, hint: none,
      min-height: auto, padding: auto)[body]
#trou-inline(width: auto)[body]

```

16.2 Setup Function

```

#beautiframe-setup(
  style: "classic",           // classic, modern, elegant, colorful, boxed,
                             // minimal, academic, bw, cours

  // — Variant mapping —————
  // default-variant sets all 7 at once; individual params override it
  default-variant: none,      // e.g. "standard" – applies to all 7 types
  theorem-variant: "prominent",
  definition-variant: "standard",
  lemma-variant: "standard",
  proposition-variant: "standard",
  corollary-variant: "standard",
  remark-variant: "subtle",
  example-variant: "accent",

  // — Colors —————
  primary-color: rgb("#2c3e50"),
  secondary-color: rgb("#7f8c8d"),
  accent-color: rgb("#2980b9"),
  background-color: white,
  // Per-environment colors (used by colorful style and accent variants)
  theorem-color: rgb("#c0392b"),
  definition-color: rgb("#2980b9"),
  lemma-color: rgb("#8e44ad"),
  proposition-color: rgb("#8e44ad"),

```

```

corollary-color: rgb("#d35400"),
remark-color: rgb("#7f8c8d"),
example-color: rgb("#27ae60"),
env-colors: false,           // true = every variant follows the env colors
label-color: auto,           // auto | "base" | a color – ink of the header
background-tint: auto,       // auto = perceptual, or a ratio like 92%
background-lightness: 0.93,   // target lightness of the tints, 0..1

// — Typography —————
label-size: 1em,             // 1em = body font size (recommended default)
label-weight: "bold",        // "bold", "regular", "semibold", etc.
name-style: "italic",        // subtitle style: "italic" or "normal"
body-size: none,            // inherited from document when none

// — Vertical spacing —————
theorem-above: 1em,
theorem-below: 0.8em,
definition-above: 1em,
definition-below: 0.8em,
lemma-above: 0.8em,
lemma-below: 0.6em,
proposition-above: 0.8em,
proposition-below: 0.6em,
corollary-above: 0.8em,
corollary-below: 0.6em,
remark-above: 0.6em,
remark-below: 0.6em,
example-above: 0.8em,
example-below: 0.8em,
proof-above: 0.5em,
proof-below: 0.8em,
header-gap: 0.3em,          // gap between label and body

// — Layout —————
inset: (x: 0.8em, y: 0.6em), // padding inside boxes
border-width: 1pt,           // stroke thickness
border-radius: 0pt,          // rounded corners (boxed style)
line-position: 2cm,          // vertical line distance from left (classic)
label-extra: 1cm,            // label overhang into left margin (classic/cours)

// — Numbering —————
numbering-format: "1",       // "1" or "1.1" (section.number)
link-to-section: false,      // false, true (1 level) or int N: prefix with
                             // the first N heading levels ("Theorem 2.1.3")
counter-reset: "manual",     // "manual" or "section" (restart at each
                             // heading up to the link-to-section depth)

// — Labels (singular) —————
theorem-label: "Theorem",
definition-label: "Definition",
lemma-label: "Lemma",
proposition-label: "Proposition",
corollary-label: "Corollary",
remark-label: "Remark",
example-label: "Example",
proof-label: "Proof",

// — Labels (plural, used when plural: true) —————
theorem-plural: "Theorems",
definition-plural: "Definitions",
lemma-plural: "Lemmas",
proposition-plural: "Propositions",
corollary-plural: "Corollaries",
remark-plural: "Remarks",
example-plural: "Examples",

// — Header layout —————
header-layout: "label-first", // "label-first", "title-first", "prefix"
label-abbrev: false,          // demote labels to the abbreviations below
prefix-separator: ":",        // separator of the "prefix" layout
theorem-abbrev: "Thm",
definition-abbrev: "Def",

```

```

lemma-abbrev: "Lem",
proposition-abbrev: "Prop",
corollary-abbrev: "Cor",
remark-abbrev: "Rem",
example-abbrev: "Ex",
proof-abbrev: "Pf",

// — TrouS —
trou-fill: "empty",           // "empty", "lines", "grid"
trou-scale: 2.0,             // handwriting factor on the measured height
trou-line-gap: 8mm,         // ruling pitch, and the quantum "lines" snaps to
trou-frame: true,           // frame the reserved space
trou-color: luma(65%),       // ink of rules and frame (student build)
trou-padding: 0.6em,        // added to the measured height
trou-min-height: 1cm,       // floor for the reserved height
trou-max-height: none,      // cap for the reserved height (none = uncapped)
trou-hint-size: 8pt,        // size of the hint text
trou-mark-instructor: true, // flag filled content in the instructor build
trou-mark-color: none,      // none = accent-color

// — QED —
qed-symbol: sym.square.stroked,

// — Advanced —
breakable: true,            // allow page breaks within environments
color-mode: "color",       // "color", "grayscale", "bw"
instructor-mode: false,    // true = show corrections, instructor-only
                           // blocks, and the content of trous
correction-label: "Correction",
correction-renderer: none,  // (title, body) => content, or none

// — QR sidebar —
qr-renderer: none,         // url => content function, or none to disable
qr-width: 1.85cm,         // width of the QR sidebar column
)

```

16.3 Utility Functions

```

#beautiframe-reset()           // Reset all built-in counters
#beautiframe-reset-french-math() // Reset built-in + French env counters
#reset-env("Label")           // Reset a specific custom env counter

// French math presets
#preset-french-math()          // cours style, color, blue accent
#preset-french-math-bw()       // bw style, black-and-white

// Language presets
#preset-english()              // back to the built-in English labels
#preset-french()
#preset-german()
#preset-spanish()

// Color themes
#theme-ocean()
#theme-forest()
#theme-sunset()
#theme-lavender()

// QED presets
#qed-square()                 // □
#qed-filled()                 // ■
#qed-tombstone()              // ▬
#qed-cqfd()                   // CQFD
#qed-slashes()                // //
#qed-text()                   // Q.E.D.
#qed-none()                   // (none)

```